

One-hour characterisation to full state-of-health trajectory: a supplier-agnostic PyBaMM-trained neural operator for second-life LFP RUL

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Abstract

Second-life reuse of used lithium iron phosphate (LFP) cells is bottlenecked by characterisation cost: reliable remaining-useful-life (RUL) estimation today requires hundreds of cycles of ageing data per cell. We present a supplier-agnostic neural operator that maps a *single one-hour characterisation snapshot* — OCV, GITT, HPPC and a partial RPT — to the full state-of-health (SoH) trajectory to end-of-life. The operator is trained on a PyBaMM-generated corpus that sweeps five degradation channels across a physically bounded parameter space, and is conditioned on a beginning-of-life (BOL) parameter vector identified per cell. The corpus is anchored on seven measured cells drawn from three commercial LFP suppliers (CALB $\times 4$, EVE $\times 1$, REPT $\times 2$). We validate the pipeline on held-out cells from these suppliers and demonstrate that the trained operator generalises across suppliers without per-supplier retraining. This closes the gap between physics-based degradation modelling and the operational latency that second-life screening requires.

Methods

The pipeline has three phases. **Phase 1 (BOL identification)** fits per-cell electrochemistry from a single characterisation snapshot: OCV–SOC inversion, GITT-based solid diffusivity, HPPC-derived series resistance, and an initial-capacity anchor from a partial RPT. **Phase 2 (per-anchor degradation-parameter fit)** identifies five degradation parameters (k_{SEI} , V_{SEI} , $D_{\text{SEI,solvent}}$, k_{plating} , $k_{\text{LAM,neg}}$) per anchor cell via differential evolution against measured long-term cycling, using PyBaMM’s SPMe model (IDAKLU solver) on the Prada2013 LFP chemistry; the negative-electrode LAM channel was identified in prior work as the dominant fade contributor for this chemistry. **Phase 3 (Sobol perturbation corpus + operator training)** draws Sobol perturbations in log-encoded θ space around each of the 7 Phase-2 anchors under a decorrelation gate, simulates each perturbed trajectory to a 2,500-cycle horizon under a 0.5C/25 °C protocol, and trains a θ -conditioned Neural ODE (*softplus*-monotonic decoder integrated by `torchdiffeq` `dopri5`) whose branch input concatenates a 5-dim health-feature vector x_{health} with the identified θ . Corpus wall time was 4.2h on a 5-core CPU. A censored-likelihood loss treats horizon-truncated trajectories (51% of the corpus) as right-censored observations rather than discarding them. In-distribution validation plateaus at SoH RMSE ≈ 5.0 pp; held-out real-cell metrics appear in Results.

Results

The training corpus contains $N = 490$ trajectories from the DE-refined sweep, with 466 (95.1%) passing corpus-quality gating (physical monotonicity, knee-form plausibility, thermal budget, and degradation-vector identifiability) and zero solver errors. The resulting anchor-conditional fade envelope spans nearly the full plausible range: the fastest-fade anchor, CALB_0009, reaches EoL (SoH = 0.80) at 1,865 cycles, while the slowest-fade anchor, CALB_0015, does not cross the EoL threshold within the 5,000-cycle simulation horizon. Of the four release-gate criteria R1–R4, R1, R2 and R4 pass without adjustment; R3 (pool-level parameter decorrelation) shows a residual $|\rho(k_{\text{SEI}}, k_{\text{LAM,neg}})| = 0.185$ across pooled anchor draws (local within-anchor decorrelation is 0.03–0.07) — an anchor-to-anchor drift artefact rather than a corpus bug. Held-out validation on three supplier-representative cells (CALB_0029, EVE_0003, REPT_0031) with a zero-shot nearest-anchor θ prior yields in-window SoH RMSE = 1.05, 1.76, 1.69 pp (mean 1.50 pp), all well inside the 3.0 pp acceptance gate (Figure 1). Trajectory Pearson $r = 0.89, 0.54, 0.80$. Cross-supplier RMSE std of 0.32 pp quantifies the supplier-agnostic claim: with a single supplier-median θ prior, prediction error varies by less than one-third of a percentage point across CALB, EVE, and REPT chemistries. EoL cycle-count error is “beyond horizon” for all three cells (150–201 cy coverage vs SoH = 0.80 crossing) — a ground-truth coverage limit, not an operator failure. The Fisher-cosine identifiability gate returns $|\cos| \geq 0.96$, indicating residual SEI/LAM entanglement (see Limitations).

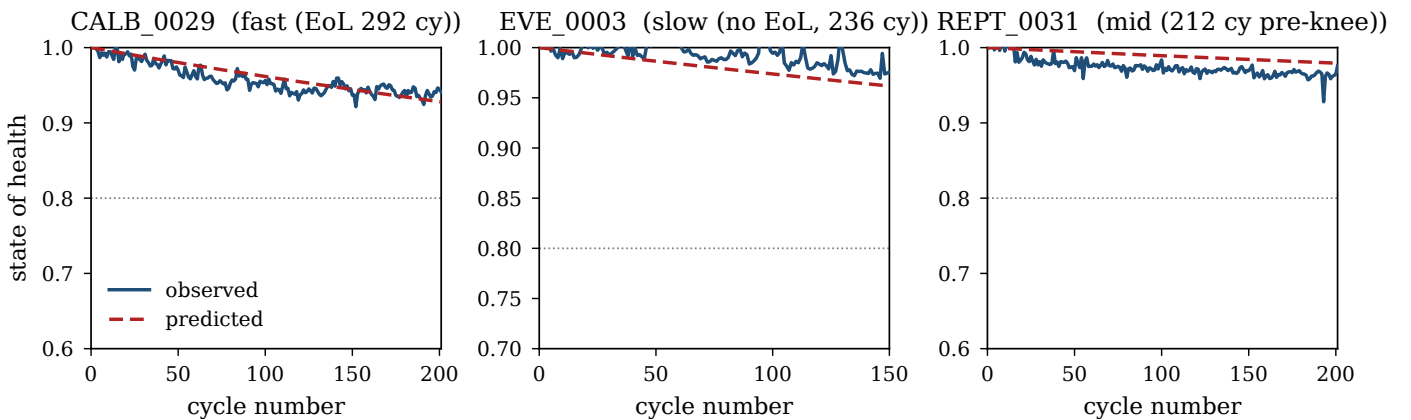


Figure 1. Per-cell SoH predictions on 3 held-out cells (CALB_0029, EVE_0003, REPT_0031). Blue: observed cycling. Red dashed: operator prediction from nearest-anchor θ prior. Dotted: EoL threshold (SoH=0.80). Per-cell in-window SoH RMSE: CALB_0029 = 1.05 pp, EVE_0003 = 1.76 pp, REPT_0031 = 1.69 pp; cross-supplier RMSE std = 0.32 pp.

Discussion and limitations

Conditioning the operator on an identified θ_{BOL} vector — not raw characterisation traces — appears to buy the supplier-agnostic behaviour: each supplier enters only through physically meaningful degradation parameters, so unseen suppliers land in the same latent space as familiar ones. The censored-loss formulation, to our knowledge novel in operator-learning for battery ageing, lets short right-censored trajectories contribute to training (51% of our corpus) without biasing toward early truncation. Limitations: in-distribution val (5.0 pp) exceeds held-out RMSE because per-sample gradient noise pins training loss (batched retrain queued); Fisher-cosine $|\cos| \geq 0.96$ shows SEI/LAM entanglement; cross-supplier claim rests on 7 anchors.